

THE KENYA NATIONAL EXAMINATION COUNCIL

KCPE 2012

TIME;2HRS

MATHEMATICS

READ THESE INSTRUCTIONS CAREFULLY

1. You have been given this question booklet and a separate answer sheet. The question booklet contains 50 questions
2. Do any necessary rough work in this booklet
3. When you have chosen your answer, mark it on the ANSWER SHEET, not in this booklet
4. Use only an ordinary pencil
5. Make sure that you have written on the answer sheet:

YOUR INDEX NUMBER

YOUR NAME

NAME OF YOUR SCHOOL

6. By drawing a dark line inside the correct numbered boxes mark your full index number [i.e. school code number and the three-figure candidate's number] in the grid near the top of the answer sheet.

7. DO not make any mark outside the boxes
8. Keep the sheet as clean as possible and do not fold it
9. For each of the questions 1-50 four answers are given. The answers are lettered A, B, C and D. In each case only ONE of the four answers is correct. Choose the correct answer
10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the one you have chosen is written.

EXAMPLE

In the question booklet:

11. What is the value of $6(24-18) + 6 \times 4$?

6

A. 30

- B. 25
- C. 10
- D. 28

The correct answer is C (10)

On the answer sheet:

1 [A] [B] [C] [D] 11. [A] [B] [C] [D] 31. [A] [B] [C] [D] 41. [A] [B] [C] [D]

In the set the box with the letter D printed in it is marked.

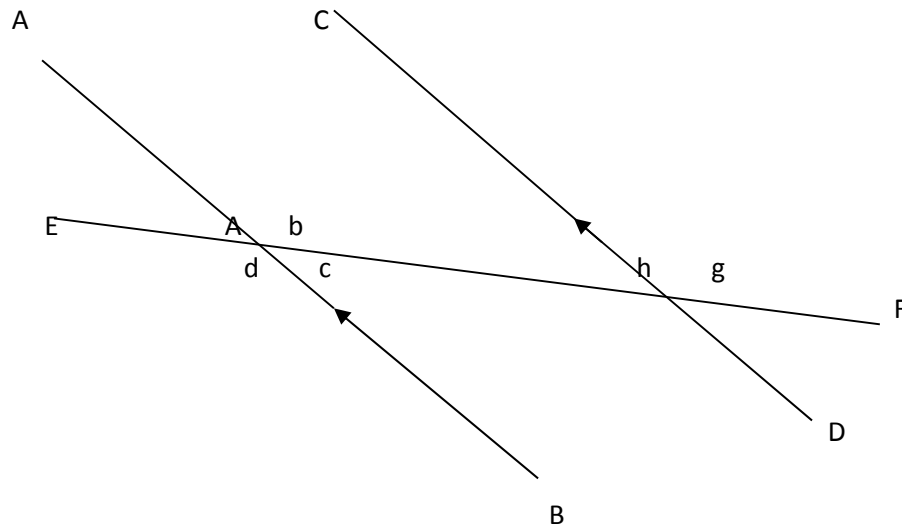
11. Your dark line must be within in the box.

12. For each quaton only one box is to be marked in each set of the four boxes.

1. What is 20075803 written on words?
 - A. Two hundred million seven hundred fifty eight thousand and three
 - B. Two hundred millions seventy five thousand egght hundred and three
 - C. Twenty millions seven hundred fifty eight hundred and three.
 - D. Twenty million seventy five thousand eight hundred and three.
2. In the number 14205, what is the difference between the total of the total value of digit 4 and 2?
 - A. 4000
 - B. 3800
 - C. 4200
 - D. 200
3. What is 4689.99975 rounded off to the nearest thousandths?
 - A. 5000
 - B. 4689.999
 - C. 4690.000
 - D. 4689.9998
4. What is the value of
 $(16905 - 1500 + 1025 - 1225) \div 5$?
 - A. 15205
 - B. 3121
 - C. 3041
 - D. 2631
5. Which of the following expression is correct?
 - A. $\frac{7}{8} > \frac{3}{4}$
 - B. $72.5 - 2.52 < 49.9 + 20.03$
 - C. $506 + 45 > 330 + 222$
 - D. $0.65 < 0.065$
6. What is the value of:
 $\frac{3}{4} + \frac{1}{8} - \frac{1}{4} \div \frac{1}{2}$?
 - A. $\frac{3}{0}$
 - B. $\frac{14}{25}$
 - C. $\frac{1}{5}$
 - D. $\frac{9}{40}$
7. What is the next number in the pattern bolow?
 2, 6, 12, 20, 30 _____
 - A. 52
 - B. 44
 - C. 42
 - D. 40
8. What is the square root of the number obtained when 196 is multiplied by 4
 - A. 28
 - B. 56
 - C. 392

D. 784

9. In the figure below line AB and CD are parallel line EF is a transversal.

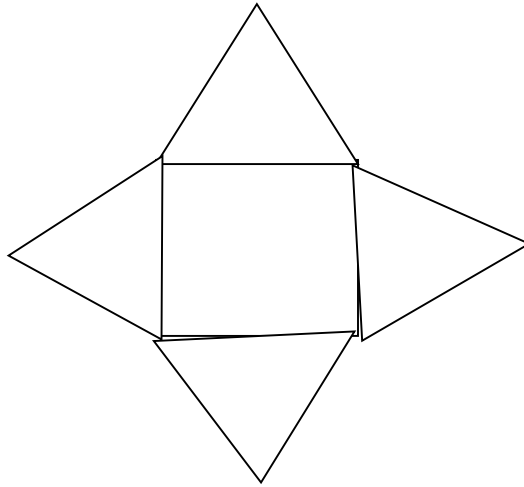


Which one of the following choices contain equal angle?

- A. A and e
 - B. C and g
 - C. B and f
 - D. D and g
10. What is the value of $2b(a + c) + ac$ When $a = b = 3$ and $c = 2$?
- A. 66
 - B. 36
 - C. 30
 - D. 21
11. A ndona bought 50 pineapples at sh 30 each and paid 100 for transport to the market. During transportation 5 pineapple got spoilt. She sold the rest at sh 50 each. What percentage profit did he make?
- A. $56 \frac{1}{4}$
 - B. $46 \frac{7}{8}$
 - C. $43 \frac{1}{3}$
 - D. $40 \frac{5}{8}$
12. Interschool soccer competition started at 3.15p.m.After 45 minute,players went for 15 minutes break. The game then took 55 minutes to end. At what time in the 24 hours system did the game end?
- A. 17 10h
 - B. 05 10 h
 - C. 16 55 h

D. 16 30 h

13. The figure below represent the net of a solid.



The net is folded to form the solid. How many edges will the solid have?

- A. 4
- B. 5
- C. 8
- D. 12

14. The table below shows the number of people who attend an agriculture show.

Males adult	Female adult	children
986	3145	5807

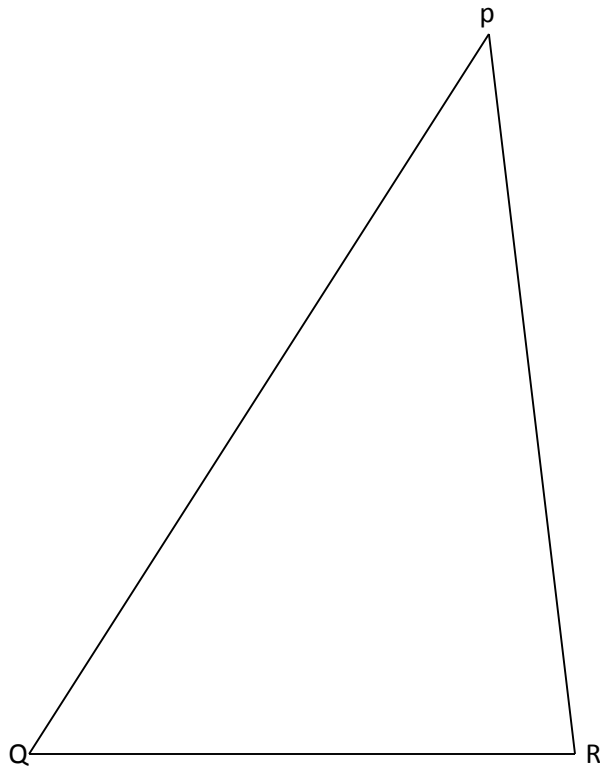
How many more children than adults attended the show?

- A. 1676
- B. 2662
- C. 4131
- D. 4821

15. A kiosk owner bought 10 litres of milk on Monday. On Tuesday and Wednesday, a total of 20 litres were bought. Equal numbers of litres were bought on Thursday and Friday. Altogether 120 litres were bought during the 5 days. In a pie chart, what angle would represent the amount of milk bought on Friday?

- A. 30°
- B. 90°
- C. 150°
- D. 180°

16. Two pickups were used to collect garbage from a market. Each pickup carried 2 tonnes 300 kg of garbage per trip. Each pickup was to make 5 trips. After the fourth trip one of the pickups broke down and did not carry the garbage. What was the total mass of the garbage collected?
 - A. 23 tonnes 0 kg
 - B. 20 tonnes 700kg
 - C. 18 tonnes 400 kg
 - D. 11 tonnes 500 kg
17. A company gives a commission on sales above sh 100 00. In a certain month a salesgirl received a commission of 36000 after selling goods worth sh 800 00. What was the percentage commission offered?
 - A. 36
 - B. 4
 - C. $4\frac{1}{2}$
 - D. $5\frac{1}{7}$
18. Paint was stored in three containers of 48 litre, 72litres and 30litres. The paint in each container was then repacked into smaller containers. The amount of paint in each of the smaller containers was the same. What was the capacity of the largest container used to repack the paint?
 - A. 3litres
 - B. 6litres
 - C. 72litres
 - D. 720litres
19. Jane gave money to her three children. She gave $\frac{1}{3}$ of the total amount to the first child and $\frac{2}{5}$ of the remainder to the second child. She was left with sh 2400 which gave to the third child. How much money did she give out altogether?
 - A. Sh 9 000
 - B. Sh 6 000
 - C. Sh 4 000
 - D. Sh 3 600
20. The mean mass of four pupils was 49.5kg. When the masses of another pupil and a teacher were included, the mean mass became 53kg. If the mass of the pupil was 16 kg less than of the teacher, what was the teacher's mass?
 - A. 44kg
 - B. 52kg
 - C. 76kg
 - D. 68kg
21. On the triangle PQR drawn below, construct line RS parallel to line QP. Draw a perpendicular from P to meet line RS at T.



What is the length of the line PT?

- A. 4.
- B. 5.1cm
- C. 6.8cm
- D. 9.5cm

22. In the year 2008 there were in a school of whom $\frac{3}{5}$ were boys. In the year of 2009, ten girls joined the school and twenty boys transferred to another school what was the ratio of boys to girls in the school, in the year of 2009?

- A. 3 : 2
- B. 53 : 33
- C. 13 : 8
- D. 7 : 5

23. A farmer harvested 900 bags of maize. She sold 0.7 of the bags and gave 0.1 of the remainder to a charitable organisation then kept the rest. How many bags of maize were kept?

- A. 27
- B. 180
- C. 243
- D. 270

24. What is the value of x in $3(x + 4) - 10 = 32$?

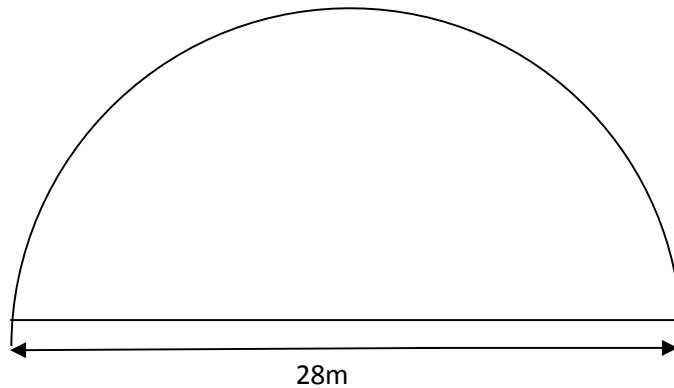
- A. $16 \frac{2}{3}$
- B. $12 \frac{2}{3}$
- C. $11 \frac{1}{3}$

- D. 10
25. In a right angled triangle the length of hypotenuse is 40cm. Which one of the following pairs are the possible length of the two shorter sides?
- A. 12cm, 5 cm
 - B. 25cm , 15cm
 - C. 24cm, 7cm
 - D. 32cm , 24cm
26. Muli obtained a total score of 180 marks in English, Mathematics and Science. His score in English is a half the score obtain in Mathematics. The score in Mathematics was 10 more than the score obtained in science. If the score obtained in science is represented by x, which one of the equation below can be used to get the value of x?
- A. $2\frac{1}{2}x + 15 = 180$
 - B. $2\frac{1}{2}x - 15 = 180$
 - C. $4x + 30 = 180$
 - D. $x + 15 = 180$
27. The table below shows inland postal charges for letters.

Weight of letters	Sh	Cts
Up to 20g	25	00
Over 20g up to 50g	30	00
Over 50g up to 100g	35	00
Over 100g up to 250g	50	00
Over 250g up to 500g	85	00
Over 500g up to 1kg	135	
Over 1kg up to 2 kg	190	00

- Wamu posted two letters each weighing 95g and another one weighing 450g. How much money did he pay at the post office?
- A. Sh 120
 - B. Sh 135
 - C. Sh 155
 - D. Sh 240
28. A trader deposited Sh 300 at a bank that paid a compound interest at the rate of 5%p.a. How much money was in her account in the bank, at the end of two years?
- A. Sh 330 750
 - B. Sh 330 000
 - C. Sh 315 000
 - D. Sh 30 750
29. On a map of scale 1: 100 000 a rectangular plot of land measures 7cm by 4cm. What are the actual length of the plot in kilometers?
- A. 70 000 by 40 000
 - B. 700 by 400

- C. 70 by 40
D. 7 by 4
30. The number of tree seedlings planted in the certain season increased by 20%. If there were 24000 seedlings planted in that season ,how many seedlings were there before increase?
- A. 19 000
B. 20 000
C. 28 800
D. 30 000
31. Which quadrilateral has all yhe properties listed below?
All sides are equal
All angles are right angles
Diagonals are equal
Diagonals bisect each other at right angles
- A. Square
B. Trapezium
C. Rectangle
D. Rhombus
32. Kioko bought the following items to donate to a charitable organization
2 bags of cabbage @ sh 25 00
4 bags of potatoes @ sh 2000
20 kg of cooking oil for shs 2000
100kg of sugar @sh 96
50 loaves of bread @ sh 35
He was given a 10% discount for all his purchases. How much did he pay for the items?
- A. Sh 23 715
B. Sh 26 350
C. Sh 57 915
D. Sh 28 985
33. Musa and Kayai left Town P for Town Q at the same time.Musa drove at an average speed of 80km/h and reached town Q after 3 hours. Kayai drove at an average speed of 50km/h for 1 ½ hrs and then continued with the journey at an Average speed of 70km/h. How many kilometers had kayai to cover at the time Musa reached town Q?
- A. 240
B. 180
C. 165
D. 60
34. A plot of land is in the shape of semicircle of diameter 28 metres as shown below.



The plot was fenced by erecting 4 metres apart. How many posts were used?

(Take $\pi \frac{22}{7}$)

- A. 12
- B. 17
- C. 18
- D. 19

35. Fifteen painter can paint a number of houses in 12 days. If the number of painters increased by 5, how many days less would it take the painters working at the same rate to paint the houses?

- A. 24
- B. 9
- C. 4
- D. 3

36. Construct triangle PQR in which PQ = 5cm, PR = 7cm, QR = 6.5cm. Bisect angle PRQ and Let the bisector meet the line PQ at M.

What is the size of angle PRM?

- A. 22°
- B. 44°
- C. 66°
- D. 95°

37. A cylindrical water tank has a diameter of 1.4m and a height of 3m. What is the valume of the tank in m^3 ?

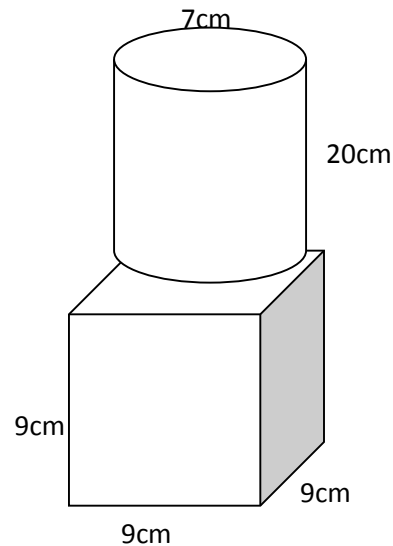
- A. 4.62
- B. 13.2
- C. 16.28
- D. 18.48

38. A baby work up at 5.30a.m. After sleeping for 7 hours 45 minutes. At what time in a.m/p.m. did the baby sleep?

- A. 9.45a.m

- B. 1.15a.m
- C. 9.45pm
- D. 2.15p.m

39. The diagram below represents a metal solid made up of a cylindrical bar fixed onto a cube. The cylindrical bar is 20cm long and has a diameter of 7 cm. Each side of the cube is 9cm long.

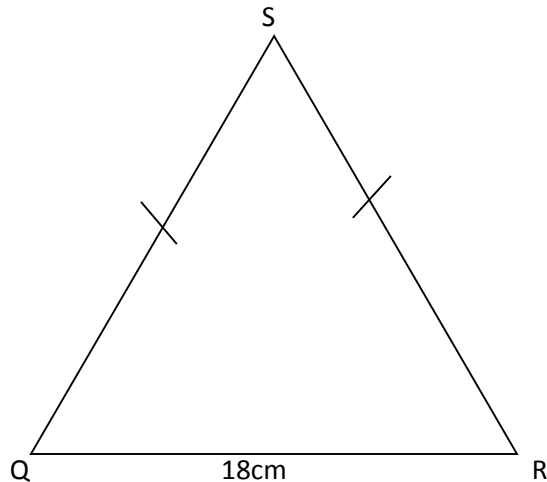


The surface of a solid was painted. What area in cm^2 was painted? (Take $\pi = 22/7$)

- A. 1499
 - B. 965.5
 - C. 926
 - D. 845
40. The table below shows the distance in kilometers from Avedi's Home to school, Health centre and market?

One day Avedi left home for the market but passed through the Healthy Center. Later Avedi left the market and went home directly. How many kilometers did Avedi travel that day?

- A. 14
 - B. 19
 - C. 18
 - D. 15
41. The perimeter of the isosceles triangle QRSShown below is 48cm .The base of the triangle is 18cm.



What is the area of the triangle?

- A. 54cm^2
 - B. 108cm^2
 - C. 135cm^2
 - D. 435cm^2
42. A factory packs water in a half litre, two litre and five litre bottles. On certain day, the factory packed 20840 bottles and 3960 were two litres bottles. One litre bottles were 2000 more than the two litres bottles. The rest were five litres bottles. How many five litre bottles were there?
- A. 2800
 - B. 6760
 - C. 6800
 - D. 18040
43. Which one of the expressions below is the simplest form of? $\frac{6(x + 2y) + 3x}{2(x + 2y) - 2y}$
- A. $\frac{9x + 12y}{2x + 2y}$
 - B. $\frac{6x + 6y}{x}$
 - C. $\frac{3 + 3x}{1 - 2y}$
 - D. $\frac{9x + 2y}{2x}$
44. At a school prize Giving day the number men was half that of women. The number of children was three times that of women. The number of women who attended was $3x$. What was the sum of men and children?
- A. $10\frac{1}{2}x$
 - B. $2\frac{1}{2}x$
 - C. $13\frac{1}{2}x$
 - D. $21x$

45. The cash prize of a radio is Sh 8000. The hire purchase price is 50% more than the cash price. Amina bought a radio on hire purchase by paying a deposit of Sh 2400 and equal monthly instalments of 800. In how months did she pay the instalment?

A. 18
B. 15
C. 12
D. 10

46. Mumbi paid Sh 10% discount was allowed. How much would she have paid for the chair had she been allowed a 12% discount?

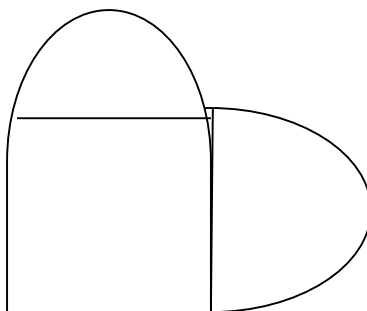
A. Sh 2000
B. Sh 1760
C. 1584
D. 240

47. The graph below shows Johns journey from Kericho to Mombasa. On his way he stopped in two places.

What was Johns average speed in Km/h between the time he left and time he arrived at the second stop?

A. 75
B. $74 \frac{2}{7}$
C. $71 \frac{7}{23}$
D. 52

48. The diagram below represents a vegetable garden which is the shape of a rectangle and a semi circle.



What is the area of the garden in square metres? (Take $\pi = 22/7$)

- A. 6933
- B. 1008
- C. 1085
- D. 1162

49. Safi packed 15 cartons each containing 20 bottles of juice. The amount of juice in each bottle was 500ml. What was the total amount of juice in litres packed by safi?

- A. 150
- B. 1500
- C. 15 000
- D. 150 000

50. Which one of the nets shown below is the net of closed cube?